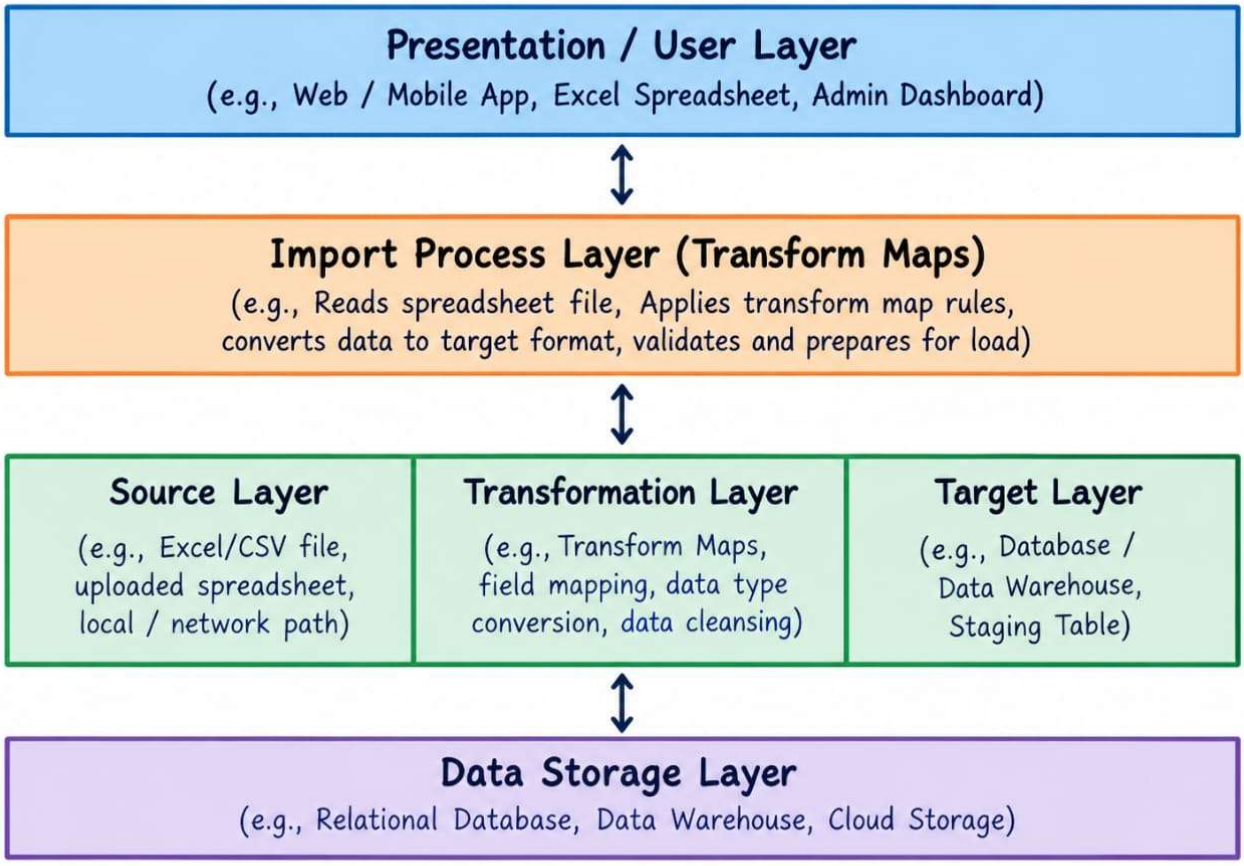
Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-8642
Project Name	Import Data Using Transform Maps (Spreadsheet)
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	Provides a user-friendly interface for uploading Excel/CSV files, configuring column-to-field mappings, triggering transformations, and viewing status logs.	React.js, HTML5/CSS3, JavaScript, Bootstrap / Tailwind CSS
[API Gateway]	Acts as the single entry point for client requests, managing route requests, rate limiting, and forwarding file streams to backend services.	Nginx / FastAPI Router / Express Gateway
[Auth Service]	Handles user authentication and role-based authorization (Admin, Data Analyst, Viewer) to ensure secure file uploads and system configuration access	JWT (JSON Web Tokens), OAuth 2.0, Passport.js
[Core Logic Service (Microservices)]	Executes spreadsheet parsing, runs transformation scripts, checks Coalesce rules to block duplicates, and maps raw records into target system schemas.	Python (Django / FastAPI) / Node.js, Pandas, OpenPyXL
[External Services / Integrations]	Handles temporary cloud storage for large spreadsheet files and sends automated email notifications for import set completions or failures	AWS S3 Bucket, Nodemailer / SendGrid API
[Database (Data Layer)]	Stores raw data in temporary Import Set staging tables, maintains Transform Map configurations, updates target application records, and logs execution errors	PostgreSQL / MySQL / MongoDB